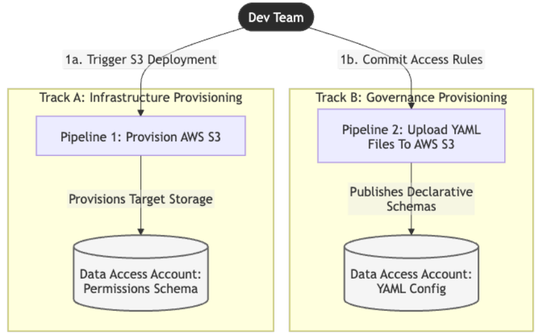
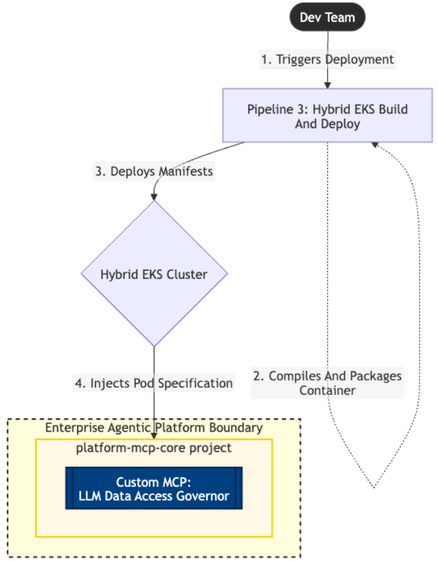
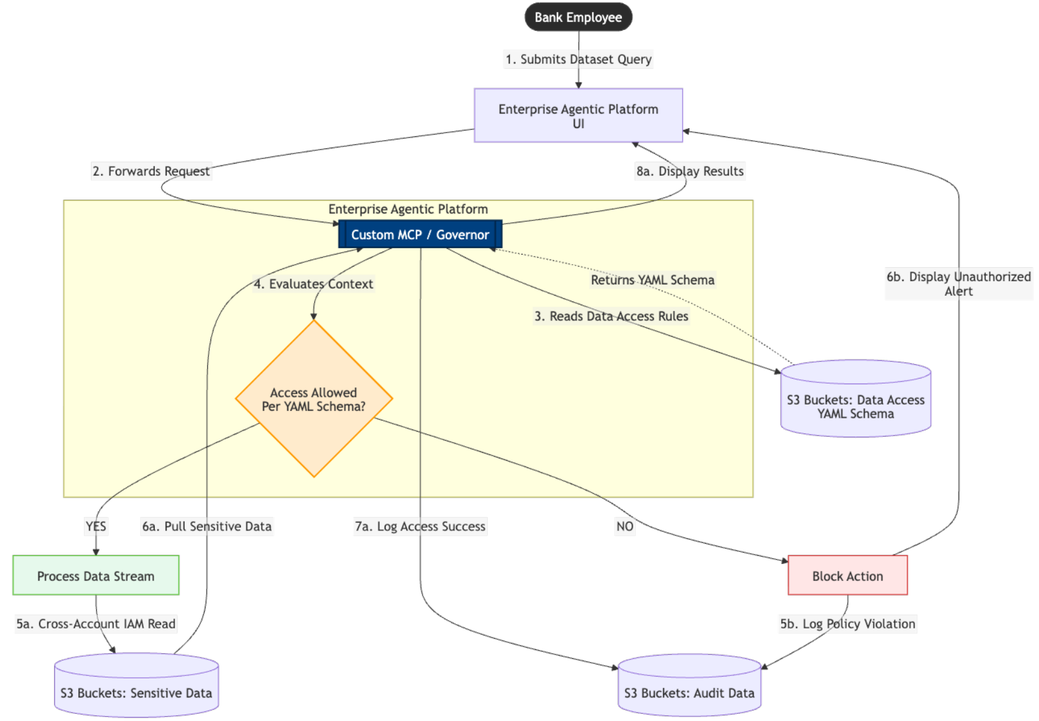
Flowchart - Setup And Provisioning (The Foundation)



Flowchart - Platform Deployment (The Engine)



Flowchart - Runtime Execution (The Data Control Flow)



LEGEND

Inbound Traffic	⟶
Outbound Traffic	- - - - - - - - - ⟶

High-Level Architecture Diagram: Enterprise Agentic Platform

Red Hat OpenShift Cluster

Application Workspace Layer

Functional Description: Houses the dedicated proxy fabric, protocol translation modules, cryptographic credential processors, and security validation layers needed to expose, secure, and monitor large language model backends, both local on-prem environments and external 3rd-party APIs.

Functional Description: Acts as the centralized, multi-tenant proxy and service discovery hub for the Model Context Protocol (MCP) ecosystem. It intercepts JSON-RPC protocol requests from agents, validates tool execution privileges against allow-lists, maps virtual endpoints to distributed microservice servers, and translates communication formats: SSE, WebSockets, and stdio, to orchestrate data ingestion.

Inbound Agent/User Request API Stream

Project/Namespace: platform-llm-core

Proxies Model Inference Requests

LLM API Gateway (Pod)

REST/gRPC Schema Validation & Request Ingress

LLM Auth Server (Internal Module)

Corporate OAuth2/JWT Claims Verification & Token Extraction

Cedar Policy Engine (Internal Module) — **Evaluates Local Cedar Policy AST** → Local Policy Store

Cedar Policy Evaluation, SafeAI Guardrails & Context Checking

LLM Router (Internal Module)

Model Routing Logic & Context Assembly

LLM Backend Connector (Internal Module)

Target Internal Proprietary LLM Models

Egress Control Firewall (Public Network)

Target External 3rd-Party LLM Models

Project/Namespace: platform-shared-control

Agent Gateway Pod

The stateless Agent Gateway intercepts client payloads, evaluates agent clearance against the Registry, and routes traffic directly across namespace boundaries to either the LLM or MCP Gateway Services based on request intent

Intra-Mesh ServiceAccount Auth & Policy Validation — **Governance Lookup & Queries Agent Card**

Agent Registry Pod

Reads Agent Capabilities, Policies & Governance

Local Agent Card Storage — JSON Metadata Describing Agent Capabilities, Policies & Governance

Asynchronous OpenTelemetry Trace Export (gRPC/HTTP) — **Asynchronous OpenTelemetry Trace Export (gRPC/HTTP)**

Agent Observability Pod

Persists Structured Spans & Execution Metrics

Metadata Storage

Trace Storage

Project/Namespace: platform-mcp-core

Proxies Context Protocol Requests

MCP API Gateway (Pod)

REST/gRPC Schema Validation & Request Ingress

MCP Auth Server (Internal Module)

Corporate OAuth2/JWT Claims Verification & Token Extraction

MCP Registry (Internal Module) — **Dynamic Tool, Prompt, & Resource Schema Cataloging** → Local Service Registry Store (ConfigMap JSON/PVC Storage)

Queries Target Server Connection Strings & Pre-Cached Tool Manifests

MCP Policy (Internal Module) — **Dynamic Session Entitlements & ACLs** → Local MCP Policy Store (Cedar / OPA Engine File)

Evaluates Tool Execution Scopes & Fine-Grained Parameter Whitelists

MCP Router (Internal Module)

Multiplexed Endpoint Target Dispatch & Logical Session Mapping

Protocol Adapters

MCP SSE Adapter (Internal Module) — **HTTP POST / SSE (JSON-RPC Web Stream)**

MCP WebSocket Adapter (Internal Module) — **Stateful Bidirectional WebSocket Stream**

MCP Stdio Adapter (Internal Module) — **Local Subprocess Standard I/O Pipes**

Mounts ConfigMap Containing Mapping Schema Templates (MCP JSON Schema —> Native Enterprise Communication Formats)

Egress Router (Private Network)

Enterprise DB

Stateful SaaS Sync

Local File Utility

Platform Engine Layer

OpenShift Routes / Ingress Controller

Endpoint Discovery & Dynamic Ingress Topology Sync

Control Node

Internal OAuth Server — **Validates Tokens & Injects Identity Claims** → K8s API Server ← **CRD Registration & Desired State Enforcement** — Operator Lifecycle Manager - OLM

Raft Consensus State Persistence (Read/Write)

ETCD State Store

High-Availability Management & Orchestration Controler: Serves as the centralized administrative authority and cryptographic trust boundary for the entire OpenShift environment. It is responsible for intercepting cluster-wide traffic, validating consumer identities, persisting infrastructure states, and coordinating global pod scheduling

Long-Polling Service Watch (Service & Endpoint Topology Stream)

Biametrical Control Sync (State Enforcement & Node Heartbeats)

Worker Node (1 of N)

Kube Proxy

Kubelet

CRI gRPC API Execution (Unix Domain Socket Directive)

CRI-O Container Runtime

Replicated Worker Node Compute: Represents the standard, horizontally scaling enterprise machine profile responsible for physically executing application workloads. It isolates low-level networking, control orchestration, and process runtimes onto individual host nodes

OCI Container Runtime Call (Namespace & Cgroup Isolation)

Red Hat Enterprise Linux CoreOS (RHCOS)

The immutable, container-optimized host operating system

Hardware Drivers (CPU/GPU)

Execute Sequential Commands

Execute Parallel Commands

On-Prem Infrastructure

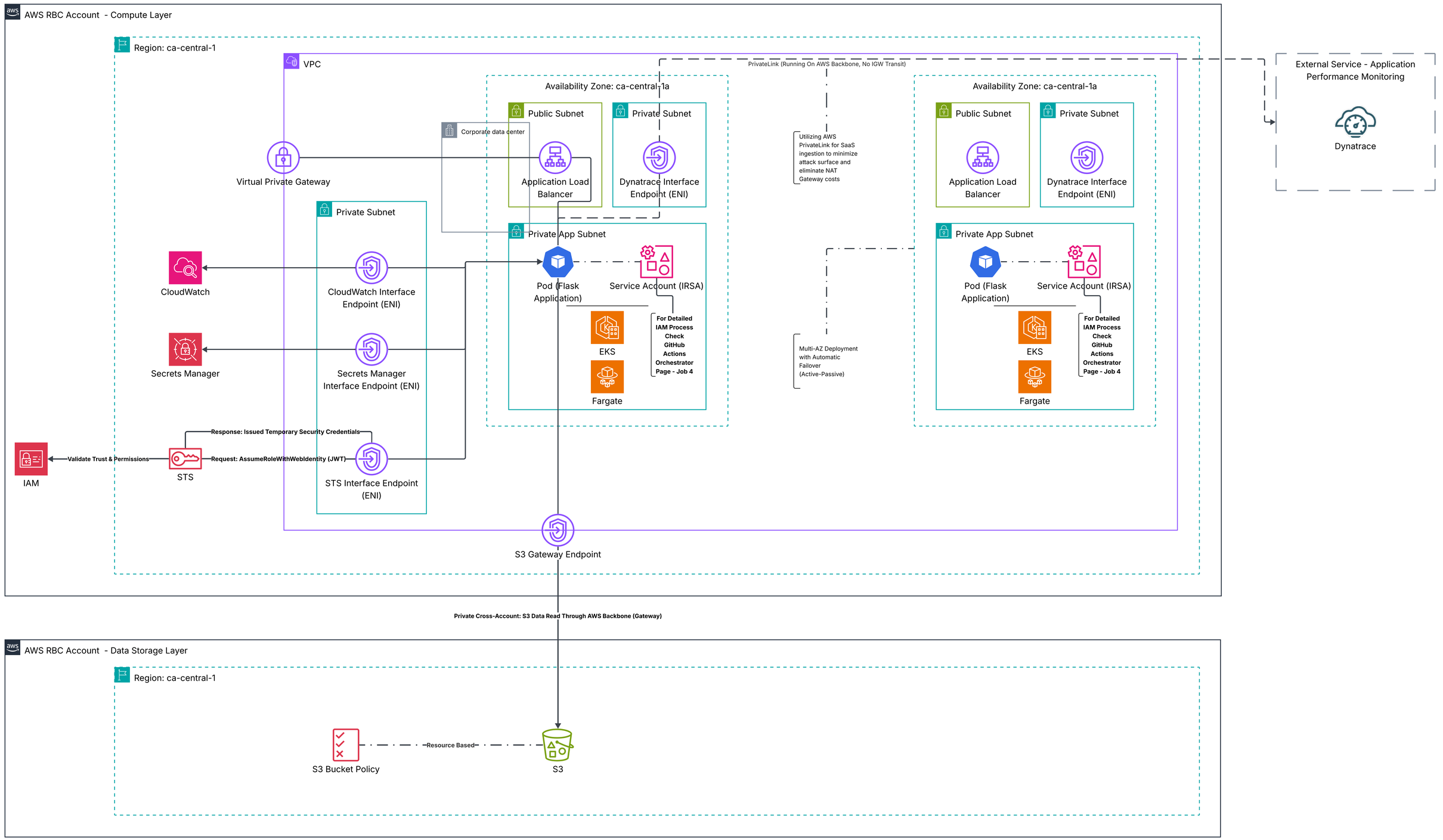
CPU Pool

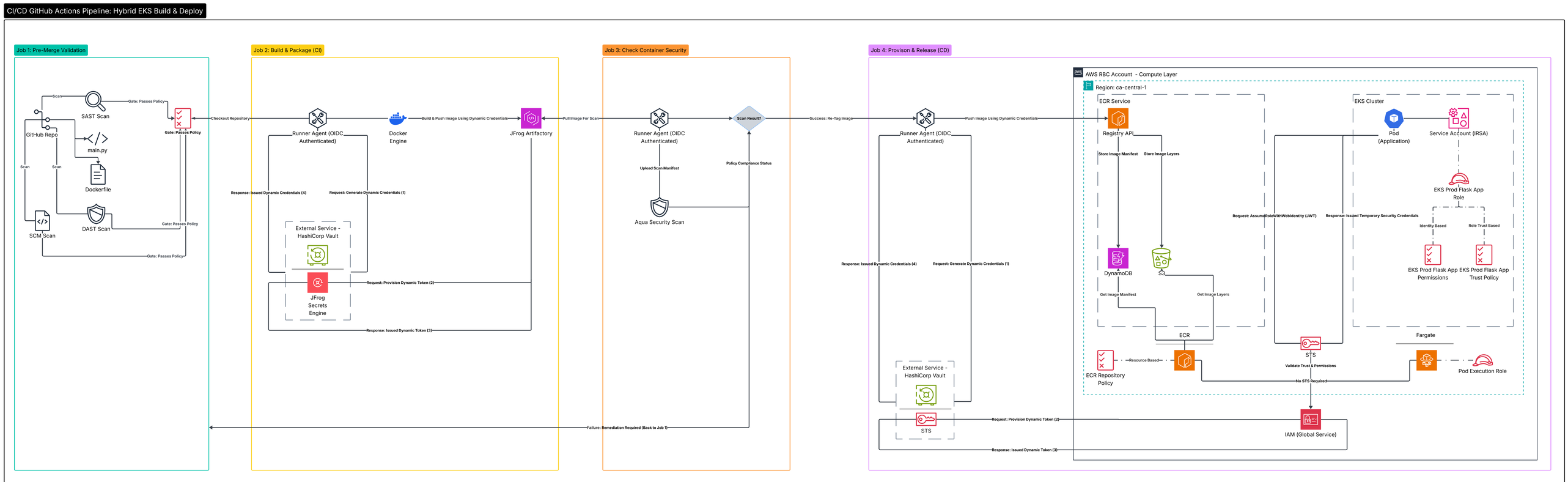
GPU Pool

LEGEND

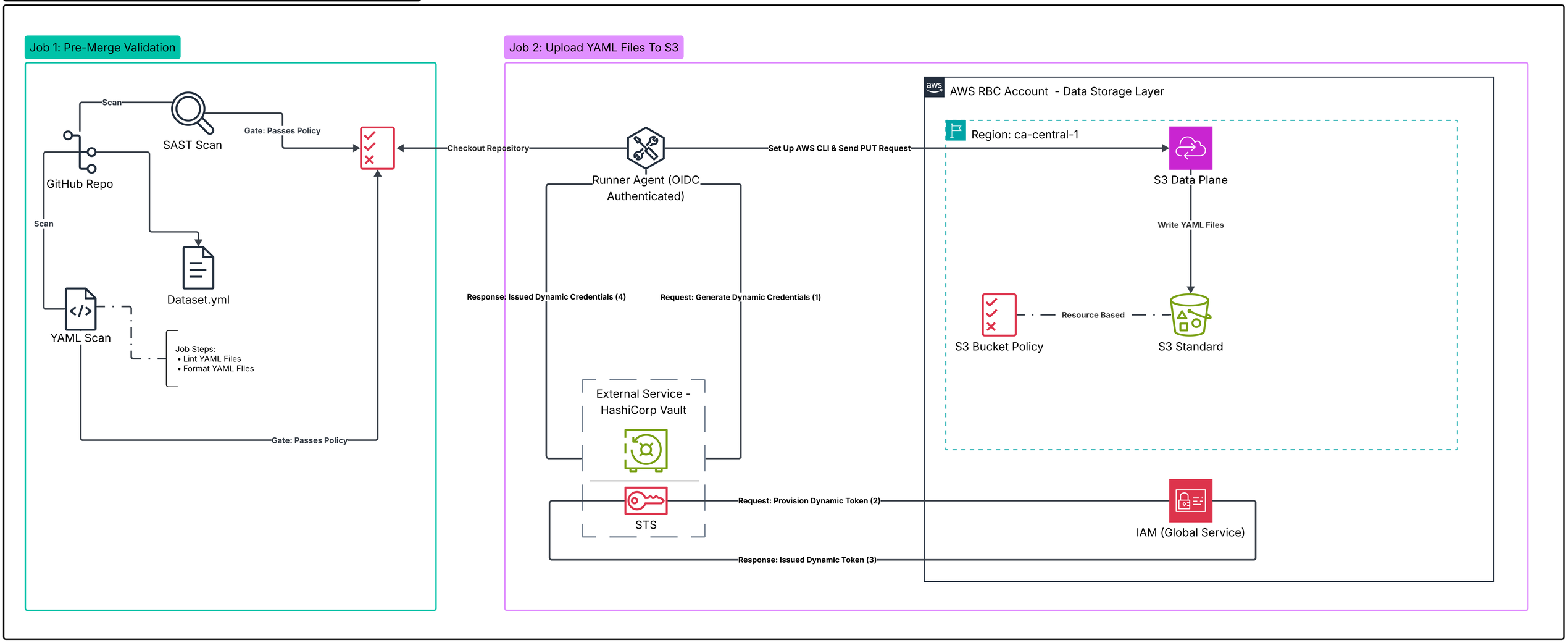
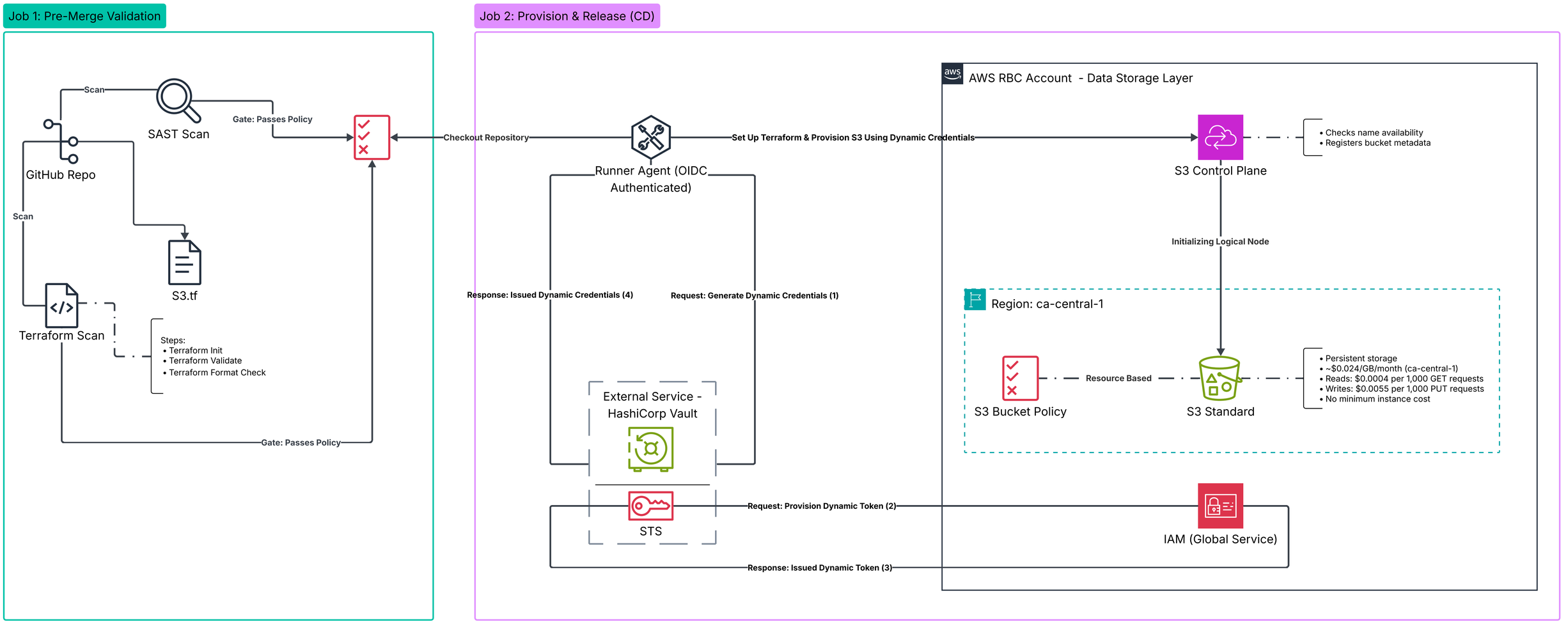
Inbound Traffic	→
Outbound Traffic	- - - →

AWS Compute & Storage Runtime: Custom MCP




CI/CD GitHub Actions Pipeline: Hybrid EKS Build & Deploy
Job 1: Pre-Merge Validation
Job 2: Build & Package (CI)
Job 3: Check Container Security
Job 4: Provison & Release (CD)
GitHub Repo
Scan
SAST Scan
Gate: Passes Policy
main.py
Dockerfile
SCM Scan
DAST Scan
Gate: Passes Policy
Checkout Repository
Runner Agent (OIDC Authenticated)
Docker Engine
Build & Push Image Using Dynamic Credentials
JFrog Artifactory
Pull Image For Scan
Response: Issued Dynamic Credentials (4)
Request: Generate Dynamic Credentials (1)
External Service - HashiCorp Vault
JFrog Secrets Engine
Request: Provision Dynamic Token (2)
Response: Issued Dynamic Token (3)
Runner Agent (OIDC Authenticated)
Upload Scan Manifest
Aqua Security Scan
Scan Result?
Policy Compliance Status
Failure: Remediation Required (Back to Job 1)
Success: Re-Tag Image
Runner Agent (OIDC Authenticated)
Push Image Using Dynamic Credentials
Response: Issued Dynamic Credentials (4)
Request: Generate Dynamic Credentials (1)
External Service - HashiCorp Vault
STS
Request: Provision Dynamic Token (2)
Response: Issued Dynamic Token (3)
AWS RBC Account - Compute Layer
Region: ca-central-1
ECR Service
Registry API
Store Image Manifest
Store Image Layers
DynamoDB
S3
Get Image Manifest
Get Image Layers
ECR
ECR Repository Policy
Resource Based
EKS Cluster
Pod (Application)
Service Account (IRSA)
EKS Prod Flask App Role
Identity Based
Role Trust Based
EKS Prod Flask App Permissions
EKS Prod Flask App Trust Policy
Request: AssumeRoleWithWebIdentity (JWT)
Response: Issued Temporary Security Credentials
STS
Validate Trust & Permissions
Fargate
Pod Execution Role
No STS Required
IAM (Global Service)

CI/CD GitHub Actions Pipeline: Upload YAML Files To S3



CI/CD GitHub Actions Pipeline: Provision S3